

AfriMentor AI: A Personality-Aligned, RLHF-Tuned Language Model Mentor for Low-Income African Youth Entrepreneurs

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Abstract

This is a placeholder abstract. AfriMentor AI is a smartphone-first, voice-note-driven mentorship platform that pairs African micro-entrepreneurs with an AI mentor persona whose reasoning style is constrained to match a documented archetype of real African achievers, trained/aligned via personality-conditioned RLHF, and grounded in an Africa-authored knowledge corpus. **[Expand once Sections 3-4 exist; state the two research questions and headline findings explicitly.]**

1 Introduction

Financial mentorship is one of the strongest predictors of small-business survival, yet it remains one of the least accessible resources for young entrepreneurs across much of Africa. Formal mentorship programs are concentrated in a handful of urban hubs, typically require an existing professional network to access, and are rarely available in the languages, financial contexts, or business categories most relevant to informal-sector entrepreneurs – market traders, smallholder agribusiness owners, and micro-manufacturers who make up a substantial share of the continent’s economic activity. At the same time, smartphone penetration across the region continues to rise even where broadband infrastructure and formal financial education do not, creating an opening for AI-mediated mentorship to reach entrepreneurs that traditional programs cannot.

Large language models (LLMs) are an increasingly plausible vehicle for this kind of mentorship at scale, but a generic conversational assistant is a poor substitute for a mentor. Mentorship is not simply information delivery; it is guidance filtered through a specific, credible point of view – the way a mentor who has personally built something from constrained circumstances reasons about risk, setbacks, and resource allocation is qualitatively different from, and more persuasive than, an undifferentiated helpful-assistant voice. This motivates the central design question behind AfriMentor AI: *can a language model be constrained to consistently reason in the style of a specific, real archetype of achievement, in a way that is both measurably faithful to that archetype and genuinely useful to the person receiving guidance from it?*

We approach this question through two complementary research threads, which this paper reports on jointly. The first (Proposal 1) is methodological: we investigate whether reinforcement learning from human feedback (RLHF), conditioned on structured personality-trait profiles derived from documented African achievers, can produce a language model whose reasoning style remains consistent across a full mentorship conversation – not just on a single prompt – and whether such a model’s outputs are distinguishable, in blind evaluation, from a generic same-capability baseline. This connects to a growing body of work on both scaling feedback signals in RLHF pipelines [Lee et al., 2024] and on measuring and shaping personality-like traits in LLM outputs [Serapio-García et al., 2025], but applies both specifically to the constraint of fidelity

to a *real, individual* archetype rather than a synthetic or aggregate persona – closer in spirit to recent work on grounding generative agents in real individuals’ documented behavior and preferences [Park et al., 2024] than to open-ended role-play.

The second thread (Proposal 2) is empirical and user-centered: what do African micro-entrepreneurs, specifically those with limited formal education and constrained access to data and devices, actually want from an AI mentorship product, and does a personality-constrained persona change engagement or perceived usefulness relative to a generic assistant? This question sits within a broader literature on LLM-based tutoring and educational support systems, much of which has been developed and evaluated in high-resource, formal-education contexts [Han et al., 2024]. We are specifically interested in whether those findings transfer to an informal-economy, low-bandwidth, voice-first context, or whether they need to be substantially rethought.

The system built to support this research, AfriMentor AI, is a smartphone-first mentorship platform with three defining constraints, chosen specifically to make both research threads testable rather than purely aspirational: (1) a knowledge base restricted to African-authored financial and entrepreneurship content, so that guidance is contextually grounded rather than generically Western; (2) a voice-note-first interaction model, chosen to match the primary user’s actual device and connectivity constraints rather than assuming typed, always-online interaction; and (3) an explicit, inspectable mapping from a small set of achiever archetypes to persona-conditioning parameters, so that persona fidelity is something we can actually measure rather than only claim.

The remainder of this paper is organized as follows. Section 2 situates this work relative to prior research on RLHF alignment, personality modeling in LLMs, generative agents, and LLM-based tutoring. Section 3 (forthcoming) describes the persona-conditioning and RLHF methodology in detail. Section 4 (forthcoming) reports the persona-fidelity and user-study evaluation. We close with a discussion of limitations specific to deploying this kind of system in low-resource, informal-economy contexts.

2 Related Work

We situate AfriMentor AI relative to three bodies of work: scaling feedback signals for RLHF alignment, measuring and shaping personality-like traits in language models, and LLM-based tutoring and mentorship systems. **A note on scope:** two citations referenced in this project’s planning materials (Han et al. and Liu et al., both nominally 2025) could not be independently verified against a specific primary source during drafting – see the flagged placeholder entries in `refs.bib`. Where a claim below depends on one of those two, it is marked explicitly as pending team confirmation rather than stated as settled.

2.1 Scaling and personalizing RLHF

Reinforcement learning from human feedback has become the standard technique for aligning language model behavior with what human raters consider helpful and appropriate, but collecting high-quality human preference labels at scale is expensive and slow. Lee et al. [2024] address this directly by showing that preference labels generated by an off-the-shelf LLM (*reinforcement learning from AI feedback*, RLAIFF) can substitute for human-labeled preferences without a meaningful drop in downstream quality, across summarization and dialogue tasks, and in some safety-relevant settings actually outperforms human-labeled RLHF. This result is methodologically relevant to AfriMentor AI’s Proposal 1: if AI-generated preference signal is a viable substitute for human labeling in general alignment, it is a natural candidate for the more specialized preference signal our approach needs – distinguishing outputs that are faithful to a specific achiever archetype’s reasoning style from outputs that are merely generically “mentor-like.”

[Pending team confirmation] A second relevant thread concerns personalizing RLHF itself, rather than

personalizing only the content a model produces at inference time – moving from a single reward model that represents an averaged human preference toward reward signals that vary by user or by target persona. This is the intended role of the Liu et al. (2025) citation in the project’s original reference list; once the specific paper is confirmed, this paragraph should state its actual method and findings rather than this placeholder framing.

2.2 Personality and persona fidelity in language models

A separate line of work asks not how to align a model’s general helpfulness, but whether LLMs exhibit measurable, manipulable personality-like traits at all. Serapio-García et al. [2025] develop a psychometric methodology for administering and validating standard human personality inventories against LLM outputs, and show that with sufficiently large, instruction-tuned models, both the *measurement* of synthetic personality traits and deliberate *shaping* of those traits toward a target profile are reliable and reproducible. This is close to a prerequisite for AfriMentor AI’s persona-conditioning approach: before asking whether a model can be constrained to a specific achiever’s reasoning style, one needs evidence that personality-like traits in LLM outputs are stable and steerable enough to constrain in the first place.

Complementary to trait-level steering, Park et al. [2024] demonstrate that generative agents grounded in structured interviews with real individuals – rather than hand-authored or purely synthetic personas – can reproduce those individuals’ survey responses and expressed values with high fidelity. This is closely aligned with AfriMentor AI’s own approach of deriving persona archetypes from documented real achievers (Epic H in the project’s implementation) rather than from generic, invented mentor personalities, and suggests our fidelity-evaluation methodology (Section 4, forthcoming) should draw on comparable validation techniques – comparing model outputs against the source individual’s actual documented statements, not just against a generic quality rubric.

2.3 LLM-based tutoring and mentorship

A growing literature evaluates LLMs as tutors or instructional assistants, most of it situated in formal education settings. Han et al. [2024] evaluate an LLM-as-tutor system for English-as-a-foreign-language writing instruction, proposing pedagogically grounded criteria for assessing student-LLM interaction quality rather than relying on generic helpfulness metrics – a distinction relevant to AfriMentor AI, where “is this response useful mentorship” is a different, and harder to define, question than “is this response helpful.”

[Pending team confirmation] The project’s reference list also cites Sharma et al. (2025) on LLM-based personalized learning; secondary citations of this work (in at least one systematic review of AI in personalized learning) describe it as examining LLMs’ potential to improve student engagement, emotional support, real-time monitoring, and adaptive assessment, while raising concerns about ethics, bias, privacy, and the difficulty of authentically simulating pedagogical strategies such as Socratic questioning. If these concerns are accurately attributed once the primary source is confirmed, several apply directly to AfriMentor AI’s context: authenticity of the persona’s reasoning style is precisely what Proposal 1 attempts to measure, and privacy is a first-class concern given the platform’s research data collection from a vulnerable user population (Non-Functional Requirement 2 in the project’s requirements).

Notably absent from this literature, and part of the gap this paper aims to address, is work situating LLM mentorship specifically in low-bandwidth, informal-economy, voice-first contexts rather than formal, typed, always-online educational settings – the constraint set that most directly shapes AfriMentor AI’s system design.

3 Method

[Not yet drafted. Depends on Epic H (trait modeling) and Epic C (persona-constrained personas) reaching enough maturity to describe concretely — both are mid-sprint as of this draft.]

4 Evaluation

[Not yet drafted. Depends on the Persona Consistency Dashboard and pilot data existing.]

5 Discussion & Limitations

[Not yet drafted.]

6 Conclusion

[Not yet drafted.]

References

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